Physics 4700H: Assignment III

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Problem 1. Consider, again, the case of N tosses of a fair coin.
On PS2, you considered the multiplicity function describing the number of ways a value of $X = N_H - N_T$ can be obtained. Using the more approximate form of Stirling's approximation you showed that $\Omega(N, X)$ is approximately Gaussian when $N \gg 1$ and $|X|/N \ll 1$. However, the form you found is not normalized. If one uses the less approximate form of Stirling's approximation, one obtains

$$\Omega(N, X) \approx \sqrt{\frac{2}{\pi N}} 2^N e^{-X^2/2N}$$

Show that in this case, the sum of the probability $P(N, X) = \Omega(N, X)/2^N$ over all possible X values is one. In the limit of N is large, the sum can be converted to an integral, but do think carefully about the limits of the sum and the conversion to the integral, and articulate clearly the approximations you are making.

Solution 1. Just to write it out,

$$P(N, X) = \Omega(N, X)/2^N \approx \sqrt{\frac{2}{\pi N}} e^{-X^2/2N}.$$

We want to show that

$$1 = \sqrt{\frac{2}{\pi N}} \sum_{X=-N,-N+2,-N+4,\ldots}^{N} e^{-X^2/2N}.$$

To work with this sum we want to convert it to an integral for which there exists an analytical solution. After the discussion in office hours I recognize that you could look at the sum and, with the desire to turn it into an integral, write it as

$$\sum_{X=-N}^{N} \frac{1}{\sqrt{2\pi N}} e^{-X^2/2N} 2$$

(by pulling 2 out from the root) which is in the general form of a Riemann sum with constant horizontal step,

$$\sum_i f(x_i)\Delta x.$$

Which can be converted to an integral by, usually, shrinking Δx to be infinitesimal. Here instead we would make the approximation that Δx is small enough relative to N that it is "effectively" infinitesimal. Instead of this I want to convert the function inside the sum to a continuous probability density function, which is defined to be a function that spits out a probability per unit length. This is because I think of continuous probability as being a function giving a probability at a point, call it $p(N, X)$ (note the little p) in our case, times a little margin in length, dX. Again as discussed in office hour this is because of Rayf's example of you being unable to be exactly 5 feet tall, instead that you can be 5 feet with plus or minus a bit, that bit being dX. To do this conversion it is intuitive to me to just, because I already have the discrete function $P(N, X)$ which has "units" of probability (in the same sense that arcsin has "units" of radians), divide by a unit length. In our case this is 2 as that is the smallest step size we can take between sampled points on $P(X, N)$. Mathematically I am saying

$$p(N, X) = \frac{1}{2} P(N, X)$$

where $p(N, X)$ is now in a continuous variable and can be integrated. I recognize the awkwardness in keeping X as that continuous variable, but I also recognize that we're making the approximation already that the step size in

X, 2, is so small relative to N that it is effectively continuous. To then extract a probability from this $p(N, X)$ we multiply it by a "length". To determine what the sum of all of these probability-length pairs is we take the integral

$$I = \frac{1}{\sqrt{2\pi N}} \int_{-N}^{N} e^{-X^2/2N} dX$$

we let $u = X/\sqrt{2N} \implies \sqrt{2N}du = dX$ which transforms the integral to

$$I = \frac{\sqrt{2N}}{\sqrt{2N\pi}} \int_{-\sqrt{N/2}}^{\sqrt{N/2}} e^{-u^2} du = \frac{1}{\sqrt{\pi}} \int_{-\sqrt{N/2}}^{\sqrt{N/2}} e^{-u^2} du$$

which is half of the error function, $\text{erf}(x)$ defined as

$$\text{erf}(x) = \frac{2}{\sqrt{\pi}} \int_0^x e^{-t^2} dt.$$

So we can write

$$\begin{aligned} I &= \frac{1}{\sqrt{\pi}} \int_{-\sqrt{N/2}}^{\sqrt{N/2}} e^{-u^2} du \\ &= \frac{1}{\sqrt{\pi}} \int_{-\sqrt{N/2}}^{0} e^{-u^2} du + \frac{1}{\sqrt{\pi}} \int_0^{\sqrt{N/2}} e^{-u^2} du \\ &= \frac{1}{\sqrt{\pi}} \int_0^{\sqrt{N/2}} e^{-u^2} du - \frac{1}{\sqrt{\pi}} \int_0^{-\sqrt{N/2}} e^{-u^2} du \\ &= \frac{1}{2} \left[\text{erf}(\sqrt{N/2}) - \text{erf}(-\sqrt{N/2}) \right] \end{aligned}$$

and since the error function is odd (which can be seen by taking the substitution $t = -t$),

$$I = \text{erf}(\sqrt{N/2}).$$

Now $\text{erf}(x)$ asymptotes to 1 *very* quickly. As an example, to 512(!) bits of precision, SageMath, an arbitrary precision computer algebra system similar to Maple, gives

$$\text{erf}(12) \approx 0.99999999999999999999999999999999999999999999999999999999999999999999986437388307940957872196938434095824273 3$$

(page cutoff for comedic effect). Since we are considering large N to be of order 10^{23} I think it is sufficient to say that this qualifies for normalization. I realize I could have taken the Gaussian integral route and approximated $-N$ and N as $\pm\infty$, but to me the asymptotic behaviour of the error function is more immediately visualizeable than the "asymptotic" behaviour of an integral, and comparing something that approaches 1 to 1 is easier than comparing something that approaches $\sqrt{\pi}$ to $\sqrt{\pi}$.